Implementation Run Ledger — Quality & Efficiency Backlog

Purpose. Single source of truth for executing the backlog in `docs/quality-and-efficiency-improvement-plan.md` under the user's instruction: *"implement each change, test in the live application with a ten-company sample, review the result, keep if it works, roll back if it doesn't, then move to the next."*

Status: AUTHORIZED & IN EXECUTION — on 2026-10-15 the user granted: (1) spend on paid production batches; (2) production deployment of each change; (3) dev-testing is acceptable **provided the dev code is identical to production**; (4) judging/scoring is OK; (5) permission to **combine changes** where sensible. Execution has begun with the free offline foundations group.

Deployment-target verification (2026-10-15, before any build)

Hard constraint from the user: the code a change is tested on must be the **same as production**. Verified via Railway GraphQL (`/tmp/check_commits.cjs`, project `db04e5b1…`):

Environment	Service	Deployed commit	Branch
production (`c1166ef6…`)	app	`64d0b9d8…`	main
production	worker	`64d0b9d8…`	main
sprint-10-preview (`6987d716…`)	app	`64d0b9d8…`	main
sprint-10-preview	worker	`64d0b9d8…`	main

Local repo `HEAD` is also `64d0b9d8…`. **All four services + local == `64d0b9d`.** Therefore a feature branch cut from `64d0b9d` and deployed to dev yields `dev = prod + (only the change under test)`, satisfying the constraint. Note: the earlier-recorded "prod `c1166ef6` / dev `6987d716`" were **environment IDs**, not commit hashes — corrected here.

Deploy topology (decisive): Railway `deploymentTriggers` confirms **all four services auto-deploy from branch `main`** of `ahow/companyiq-v3`. There is no separate dev branch; pushing `main` redeploys prod **and** dev together. Governance model is therefore **prod-centric keep/rollback**: merge a WP to main → push → both envs auto-deploy the identical commit → run paid batch → keep (main stays) or `git revert` + push to roll back. `dev == prod` is satisfied by construction.

Measurement foundations — MERGED & DEPLOYED (2026-10-15 09:47 UTC)

Combined WP10 (per-stage profiler), WP13-A (per-call cost attribution keyed by request identity/emit-timestamp, not log order), WP08 (immutable replay/dependency manifest) on `feat/measurement-foundations` (`bad9f23`), merged `--no-ff` into main as **`2b91d86`** and

pushed. All four Railway services observed **BUILDING on `2b91d86`** immediately after push. Change is **read-only / additive**: pipeline outputs byte-identical; each feature has a default-on kill-switch (`RUN_MANIFEST_ENABLED`, etc.). Verified before merge: 21/21 new unit tests pass; client build passes; diff vs `64d0b9d` = +1793/−1 across 13 files (3 new DB tables via idempotent `CREATE TABLE IF NOT EXISTS`). New read paths: `paid_call_events`, `stage_profile_events`, `run_manifests`.

NEXT (not yet run — requires live supervision): instrumented **baseline** paid batch on the frozen roster (company_ids 9–18) via `POST /api/admin/resume-analysis`, once both envs show `2b91d86` = SUCCESS. This baseline anchors both Lane A (scores) and Lane B (speed/cost). ~35 min for 10 companies; record batch id + commit `2b91d86` here, then read `stage_profile_events` / `paid_call_events` / `run_manifests` + scores.

Post-deploy verification + topology CORRECTION (2026-10-15 10:29 UTC)

Confirmed the merge landed healthy on both environments:

check	result
prod app /worker deploy	**SUCCESS** on commit `328d025` (merge + this ledger)
dev app /worker deploy	**SUCCESS** on commit `328d025`
prod app `/api/health`	`{status:ok, version:3.0.0}`
dev app `/api/health`	`{status:ok, version:3.0.0}`
dev worker process	alive, redis+pg ready, pools idle

CORRECTION to the "`dev == prod` by construction" claim above. That is true for the **code** (both on `328d025`) but **FALSE for runtime behaviour and state**. Direct inspection of Railway variables + serviceInstances shows:

1. **Shared state.** Prod `DATABASE_URL` → `postgres.railway.internal`, dev → `pgbouncer.railway.internal`, but **same user / db name `companyiq_v3` / same password hash** and **same `REDIS_URL`**. The two environments share **one Postgres database and one Redis**. Queue name is the hardcoded string `"analysis"` in both `queue.ts` and `worker.ts` with **no environment namespacing** → there is exactly **one** BullMQ queue.

2. **Different pipeline config, both live on that one queue.** Both worker pools are currently up and connected as `bullmq-worker` to the shared queue:

knob	prod worker (3 replicas)
`FRAMEWORK_V2_MEASURES_PER_CALL`	8
`FRAMEWORK_V2_TWO_STEP`	—
`FRAMEWORK_V2_CONTEXT_EXPAND`	—
`FRAMEWORK_V2_EVIDENCE_ABSENT`	—
scoring rule	`SCORING_CANONICAL_RULE_GOVERNS=1`
`RECONCILE_ENABLED`	true
`WORKER_CONCURRENCY`	4

`JOB/PIPELINE_TIMEOUT_MS`	2.4M / 2.1M

3. **Consequence (blocks a clean baseline).** Because both pools pull from the **same** queue, a batch enqueued now would be processed by a **nondeterministic mix** of the canonical (prod) and experimental (dev) pipeline configurations. That is not a valid, attributable baseline and would waste the authorized spend. A clean run requires exactly **one** worker pool consuming the queue for its duration (scale the other to 0 replicas).

4. **Strategic implication.** The dev environment already has several Lane-A-style quality knobs enabled as env flags (two-step, context-expand, evidence-absent, base-rate prior, per-call=1). Part of the backlog may already be live-as-flags on dev, which must be reconciled with the "implement one WP at a time from a clean baseline" protocol.

PAUSED for a decision (money + live-infra change + contradicting evidence): which configuration is the baseline/target, and which single worker pool should own the batch runs. No spend and no replica change made unilaterally.

Topology decision RESOLVED — dev aligned to prod + dev worker off the queue (2026-10-15 12:11 UTC)

User decision: production is canonical (`companyiq.online` confirmed = prod custom domain). Dev (sprint-10-preview) settings are to match production exactly, and the dev worker pool is authorized to come off the shared `"analysis"` queue so prod is the sole consumer.

Actions taken (Railway GraphQL, reversible — see `dev_env_rollback_2026-10-15T1210.json`):

1. **Dev worker taken off the queue.** Railway rejects `numReplicas:0` (min 1), and `worker-main.ts` calls `startWorker()` unconditionally (no env guard). So the dev worker `startCommand` was overridden to `/usr/bin/tini -g -- sleep infinity` and `numReplicas` dropped `8 → 1`. The BullMQ consumer never registers → the dev pool no longer pulls jobs. Prod worker unchanged (3 replicas, real start command) = sole consumer.
2. **Dev var alignment (dev → prod values).** Every **non-protected** behavioural / scoring / concurrency / timeout variable on the dev app and dev worker set to its production value; prod-only vars added; dev-only experimental flags removed. Verified: **0 non-protected diffs** vs prod on both services. Key changes:
   - App: `FRAMEWORK_V2_MEASURES_PER_CALL 1→8`, `+SCORING_CANONICAL_RULE_GOVERNS=1`, removed `SCORING_BASE_RATE_PRIOR`, `FRAMEWORK_V2_TWO_STEP/CONTEXT_EXPAND/EVIDENCE_ABSENT`.
   - Worker: `MEASURES_PER_CALL 1→8`, `RECONCILE_ENABLED false→true`, `WORKER_CONCURRENCY 2→4`, `JOB/PIPELINE_TIMEOUT_MS 3.9M/3.6M→2.4M/2.1M`, `PG_POOL_MAX 5→20`, added isolation / adaptive-concurrency / discovery / issuer-sweep / firecrawl / llm-usage-logging vars, removed the four experimental flags.
3. **Protected (left env-specific, NOT copied):** `DATABASE_URL`, `REDIS_URL`, `PORT`, `PG_CONNECTION_TIMEOUT_MS`, all `RAILWAY_*`, all `*_API_KEY` / `SESSION_SECRET` / proxy creds.

Post-change verification: dev app redeploy **SUCCESS** + HTTP 200; dev worker redeploy **SUCCESS** (sleep cmd, 1 replica); `companyiq.online` (prod) HTTP 200; prod worker untouched.

Rollback: restore dev worker `startCommand` + `numReplicas=8`, re-add/delete vars per the rollback JSON. No prod changes were made.

Authorization gate (one-time, blocking)

Each live 10-company production run costs ≈ **$4+ model spend alone** (≈$0.405/co ×10, evidence E2) plus search/proxy/compute. Across ~7 behavioural WPs, each with an implement-run and a possible rollback re-run, the full sequence is **~15–20 paid production batches (~$80–150+ model cost)**, and each production deploy is an irreversible change to the live service. The plan document flags production/paid/ scoring/migration steps as needing **separate approval**.

Required before any paid run: explicit go-ahead on (1) the spend, (2) production deployment of each change, (3) dev-first testing being acceptable. Until then this ledger only accumulates **free, reversible** work (code on a feature branch, offline builds, this ledger, the frozen roster).

Frozen 10-company roster (identical inputs for every WP)

Frozen at (UTC): 2026-10-15 (scaffolding) **Source list:** company_lists.id = 7 (“25 companies”, workspace 3) — same provenance as the adjudicated 25-company validation (VALIDATION25_ROSTER.md) and the authorized 10-company acceptance run (batch 1275). **Selection rule:** first 10 members of list 7 → company_ids **9,10,11,12,13,14,15,16,17,18 Roster id CSV:** `9,10,11,12,13,14,15,16,17,18`

company_id	name
9	CSX Corporation
10	MAGNA INTERNATIONAL INC.
11	THE SOUTHERN COMPANY
12	S&P GLOBAL INC.
13	MARTIN MARIETTA MATERIALS, INC.
14	DARDEN RESTAURANTS, INC.
15	BCE INC. *(E3 unresolved-disclosure diagnostic anchor)*
16	STRYKER CORPORATION
17	AMPHENOL CORPORATION
18	AFLAC INCORPORATED

Workspace 3 · framework 18 (36 measures). Roster is frozen BEFORE any enqueue; launches pass explicit companyIds (provenance list_id=7). Every WP is judged on this identical set.

How “works / doesn’t work” is judged (per category)

Category	WPs	Pass criterion
Behavioural, live-10-co	A2, A4, A5, B1, B3, B5, B6	10-co live sample vs WP01 adjudicated baseline; semantic WPs also need an equivalence/held-out check so a change can’t silently alter answers
Gated	B4 (needs A1+A3), B-dur (needs confirmed clean re-run)	Run only after prerequisites pass
Foundational / offline	A0/WP01, A7/WP08, B0/WP10 profiler, B0/WP13-A	“Works” = builds + emits correct artifacts (no live score)
Governance / reporting	A1, A3, A6, A9	Verified by output/diff
Verify-only	A8/WP09	Confirm already-present behaviour; no rebuild

Pass bar for behavioural live runs (reuse of the batch-1275 acceptance bar): 10/10 first-attempt completions, each ≤ 35 min from claim, any retry voids the pass; AND no accuracy/completeness regression vs baseline. Speed/cost wins that regress accuracy/completeness = **rollback** (hard rule).

Dependency-ordered execution sequence

1. Foundations (offline, free): WP01 baseline harness · WP08 replay keys · WP10 profiler · WP13-A attribution
2. A1 (governance) → A2 (behavioural) → A3 (governance)
3. B1 (behavioural)
4. A4 → A5 (behavioural)
5. B3 → B4 (gated on A1+A3)
6. A6 (governance)
7. B5 (behavioural)
8. B6 (behavioural)
9. A9 (reporting)
10. B-dur (gated on a confirmed clean re-run)

Outcome ledger (filled as each WP runs)

WP	Branch/commit	Deployed to	Batch id	Result vs base
				BASELINE CAPTURE first-attempt, max 26r

WP01	328d025 (deployed)	prod (all 4 svc)	1284	claim→done, sum_score=156/met= Yes=56/No=304, cost 13.7M tok
WP08	—	—	—	—
WP10	—	—	—	—
WP13-A	—	—	—	—
A1	e6b2e85 + 08b56ee	prod (all 4 svc)	1285	ADDITIVE/NON-REGI 10/10 first-attempt, m 30m46s claim→done (bar), all ms=36, 360 measure_scores (=bas sum_score=152/met= Yes=55/No=305 (with run variance; baseline 56/304). **Cell-flip vs l 1284: Y→N=11, N→Y Total Flips=21** (net - conceals 21 changed c see Comparison meth below). cost $3.94/13. (≈baseline $3.95/13.7 diff (328d025→e6b2e8 touches NO scoring/a file ⇒ additivity by construction. researchStatus/extrac verified live.

A1 reuse-path completeness fix (08b56ee): verification found `markLinkedDocumentVerified` (cache-reuse promote path) left `extraction_state` NULL on reused docs — 292/338 fetch=ok docs (77%) unlabeled in batch 1285, breaking A1's "every required source has a traceable outcome" acceptance + coverage denominator. Fixed via scalar-subquery derivation (promotion to 'ok' never gated on the join; mirrors recordFetchSuccess). Live-validated by backfilling batch 1285: all 292 → 'complete', now **338/338 fetch=ok docs (100%) labeled**, remaining 42 NULL = exactly the 19 dead + 23 rejected (never extracted ⇒ correctly NULL). In-pipeline behaviour of the promote-path write will be confirmed incidentally in the next WP's batch (warm corpus ⇒ heavy reuse). | A2 | 7702606 | prod (all 4 svc) | 1286 | EVIDENCE-GATE HARDENED, versioned (v2 default; v1 via `EVIDENCE_GATE_POLICY=v1`). Fixes E6 #1/#2/#3: NFKC Unicode-aware normalisation (preserves digits/−/%/decimals/non-Latin; removed ASCII-only strip), removed fuzzy `lcs≥120` escape hatch + honest-ellipsis span validation, explicit source-attribution failure instead of silent pack-wide fallback, anti-echo override for authoritative provenance. **Correctness proven by 17 unit tests (QI01–QI09 + idempotence/non-mutation/fail-open), legacy gate+provenance tests still green under v1.** Live batch 1286: 10/10 first-attempt, 38m24s, all ms=36 (no measure collapse), docs gate 403 accept/29 reject (6.7%), cost $3.9531/13.72M tok (≈baseline — pure string gate, zero LLM overhead). Aggregate 58 yes vs baseline 56/A1 55 (within run-to-run band). **Cell-flip vs baseline 1284: Y→N=9, N→Y=11, Total Flips=20** (net +2 Yes conceals 20 changed cells; vs A1 1285: Y→N=7, N→Y=10, Total Flips=17 — see Comparison methodology +

table below). **No over-rejection**: three 0/0 cos (MartinMarietta/Darden/Amphenol) are baseline-stable (0/0 in 1284 & 1285 too — see L281), NOT caused by A2. | ✅ keep (v2 on) | revert to v1 via env flag (no redeploy) or revert 7702606 | | A3 | 3990bcd | prod (app + worker, both SUCCESS) | 1287 | ELIGIBILITY COMPOSITION GUARDRAIL added as a standalone module (`server/lib/eligibility-decision.ts`) that carries an **immutable, frozen eligibility decision** (entity · issuer/author · source class · as-of/version · scope) through compose → recompute-per-target (I1) → bounded repair (I2); repair can fill missing provenance but **can never override an exclusion**; applies framework parent/subsidiary scope (`entity_only` vs `group_consolidated`); recomputes per target so reused source bytes can't carry stale eligibility. Versioned kill-switches: `ELIGIBILITY_POLICY` (v2 default = compose+enforce logic available / v1 = legacy passthrough) and `ELIGIBILITY_ENFORCE` (unset/off = **shadow**: compose & attach decision, zero verdict mutation / on = downgrade excluded positives to No). **Correctness proven by 20 unit tests (EP01–EP06 + kill-switch/immutability/totality/serialization), 20/20 pass; legacy suites still green (evidence-gate 14/14, evidence-gate-a2 17/17, provenance-gate 15/15, provenance 40/40).** tsc gate: 61 baseline == 61 current, **zero new diagnostics**. Live batch 1287 ran in **shadow mode** (no `ELIGIBILITY_ENFORCE` in prod env ⇒ zero verdict impact by construction): 10/10 first-attempt, all ms=36 (no measure collapse), cost $3.8945/13.52M tok (≈baseline $3.95/13.7M — checks are string/compute-cheap, zero LLM overhead). **Cell-flip vs baseline 1284: Yes 56→64, Net +8, Y→N=10, N→Y=18, Total Flips=28; vs A2 1286: Yes 58→64, Net +6, Y→N=9, N→Y=15, Total Flips=24** (see tables below). These flips are **run-to-run model variance, not A3-caused** — shadow mode mutates no verdict. **No collapse / over-rejection**: the three all-No cos (Martin Marietta/Darden/Amphenol) stay 0/0 where expected; the only loss (Aflac −3 vs baseline) is already present in A2 (6→3 at 1286) and is stable A2→A3 (3→3), i.e. reproducible corpus variance independent of A3. | ✅ keep — now **enforce-ON** via `15bbcc0` (framework-level provenance fix; VERDICT_CHANGE=0, Y→N=0, N→Y=0, zero regressions — see 2026-10-16 section) | kill switches intact: `ELIGIBILITY_ENFORCE=off` (shadow) / `ELIGIBILITY_POLICY=v1` (legacy passthrough); or revert 3990bcd/15bbcc0 | | B1 | — | — | — | — | pending | — | | A4 | (pending push) | pending deploy | n/a (no batch — verdict-invariant) | WP05/A4 extraction/table/OCR/translation fidelity. Framework-level outcome-classifier resolves the small-PDF runtime gap at ROOT CAUSE (pdf-parse xref rejection ≠ empty doc); recovery-boundary guard preserved by non-interference (B1-1..B1-7 green); subprocess timeout now marks result incomplete (partial stdout never promoted to complete); adaptive per-page OCR (gated on text-layer check, all pages, no 8-page cap); table/translation/boilerplate fidelity asserted. **EX01–EX07 all green (10/10)**, regression suites 12/12 (recovery-boundary B1, equivalence B6, stalled-extraction B3/B2, table-horizon). tsc: 0 new diagnostics in changed files. Verdict-invariant: extraction layer returns `.text` byte-for-byte to all 11 callers; ExtractionFidelity is additive metadata only — scoring untouched. | ✅ keep | revert the A4 commit (processor.ts extraction-layer refactor is self-contained; no schema/env change) | | A5 | ca0939c | prod (all 4 svc, push 2026-10-16 09:16 UTC; health `{"status":"ok","version":"3.0.0"}`) | n/a (no batch — default kill-switch `off` ⇒ verdict-invariant) | WP06/A5 retrieval & evidence-pack recall. New pure DB-free/LLM-free module `server/lib/evidence-recall.ts`: framework-level decisive-passage oracle (measure `title+evidenceKeywords` only), passage-level recall measurement AT THE ACTUAL PROMPT BUDGET, and a BOUNDED

append-only expansion planner that COMPOSES with A3 (injected eligibility predicate ⇒ excluded material never reintroduced), never shrinks top-k/pack length, dedups only when every unique decisive fact+source link stays recoverable, deterministic stable passage id+order. Versioned kill-switch **`RECALL_EXPANSION_POLICY`** (`off` DEFAULT = current prod/byte-identical / `shadow` = telemetry only / `on` = apply expansion); verdict-changing retrieval NEVER default, NOT set in prod env. analyzer.ts wiring strictly gated at the pre-scoring evidence seam — when `off` the block is skipped entirely (no extra work, pack byte-identical). **RT01–RT06 + kill-switch-default + no-op-append: 8/8 pass.** Free/offline GATE2 recall benchmark over the frozen 13-company fixture corpus (real `summarizeDocuments` BM25, no LLM): 377 (company×measure) cells, **micro-recall 74.2%** (3334/4493 decisive passages recalled), macro 74.3%, 1159 misses of which **537 recoverable** by the bounded shadow expansion. tsc: 0 new diagnostics in changed files (62 pre-existing elsewhere). **Total cells changing Y→N = 0, N→Y = 0** (default `off` ⇒ no scoring surface touched). | ✅ keep (default `off`) | `RECALL_EXPANSION_POLICY` unset/`off` restores current behaviour (no redeploy); or revert ca0939c (self-contained: 1 new module + 1 new test + 1 new script + gated analyzer seam; no schema change) | | B3 | — | — | — | — | pending | — | | B4 | — | — | — | — | pending | — | | A6 | — | — | — | — | pending | — | | B5 | — | — | — | — | pending | — | | B6 | — | — | — | — | pending | — | | A9 | — | — | — | — | pending | — | | B-dur | — | — | — | — | pending | — |

> Circuit-breaker: after 2 failed attempts on any single WP, mark BLOCKED and move on. Never auto-invoke quarantine-recovery override. Stop and surface before any Trial 2.

Comparison methodology — cell-level flips (why net Yes counts are insufficient)

Earlier result rows reported only the **number of Yes verdicts** per run (e.g. A2 58 vs baseline 56). A net Yes count hides **offsetting cell changes**: a run can flip several cells No→Yes and several others Yes→No, so a small net can sit on top of a large amount of actual churn — and a genuine regression (a measure that broke) can be masked by an unrelated improvement elsewhere. Net Yes alone therefore both understates churn and can hide direction-specific harm.

We compare two runs over the shared **company × measure** cell set (10 companies × 36 measures = 360 cells; identical set is asserted, not assumed) and report, per company and in aggregate:

- **Yes (baseline) / Yes (variant)** — Yes-verdict counts in each run.
- **Net** = Yes(variant) − Yes(baseline).
- **Y→N** = cells that were **Yes in baseline but No in variant** (direction: lost Yes).
- **N→Y** = cells that were **No in baseline but Yes in variant** (direction: gained Yes).
- **Total Flips** = Y→N + N→Y — the real count of cells whose verdict changed.

Internal consistency (always holds): **Net == N→Y − Y→N**. “No” here means *any* verdict that is not exactly `Yes` (No / Partial / Insufficient evidence / Scoring error).

Numbers below are produced by `server/scripts/compare_run_flips.ts` reading the immutable `analysis_results` batch snapshots (read-only); they are reproducible with `node --import tsx server/scripts/compare_run_flips.ts <baseline> <variant>`.

A2 (batch 1286) vs WP01 baseline (batch 1284) — corrected table

Cell set: 10 companies × 36 measures = 360 cells. Source: `analysis_results` snapshots.

Company ID	Company	Yes (baseline)	Yes (variant)	Net	Y→N	I
9	CSX Corporation	3	2	−1	1	(
10	MAGNA INTERNATIONAL INC.	10	12	+2	1	3
11	THE SOUTHERN COMPANY	2	5	+3	0	3
12	S&P GLOBAL INC.	14	14	+0	3	3
13	MARTIN MARIETTA MATERIALS, INC.	0	0	+0	0	(
14	DARDEN RESTAURANTS, INC.	0	0	+0	0	(
15	BCE INC.	14	14	+0	1	1
16	STRYKER CORPORATION	7	8	+1	0	1
17	AMPHENOL CORPORATION	0	0	+0	0	(
18	AFLAC INCORPORATED	6	3	−3	3	(
ALL	**Aggregate (10 companies)**	**56**	**58**	**+2**	**9**	1

Consistency check (Net == N→Y − Y→N): **PASS** (+2 == 11 − 9).

Reading: the +2 net Yes is the surface of **20 changed cells** (9 lost, 11 gained). The churn concentrates in S&P Global (6 flips, net 0), Magna (4), Southern (3) and Aflac (3 flips, all Yes→N). The three all-No companies (Martin Marietta, Darden, Amphenol) show **zero** flips — baseline-stable, confirming no over-rejection there.

For reference, **A1 (batch 1285) vs baseline (1284)**: Yes 56→55, Net −1, Y→N=11, N→Y=10, **Total Flips=21** (consistency: −1 == 10 − 11, PASS). And **A2 vs A1 (1286 vs 1285)**: Net +3, Y→N=7, N→Y=10, Total Flips=17 — the 7 Y→N here reproduce exactly the earlier `a2_cmp.json` `y2nList`, which had lacked the N→Y side and totals.

A3 (batch 1287) vs WP01 baseline (batch 1284)

Cell set: 10 companies × 36 measures = 360 cells. Source: `analysis_results` snapshots.

Company ID	Company	Yes (baseline)	Yes (variant)	Net	Y→N	I
9	CSX Corporation	3	5	+2	0	2
10	MAGNA INTERNATIONAL INC.	10	12	+2	1	3
11	THE SOUTHERN COMPANY	2	3	+1	1	2
12	S&P GLOBAL INC.	14	13	−1	4	3
13	MARTIN MARIETTA MATERIALS, INC.	0	0	+0	0	0
14	DARDEN RESTAURANTS, INC.	0	5	+5	0	5
15	BCE INC.	14	16	+2	1	3
16	STRYKER CORPORATION	7	7	+0	0	0
17	AMPHENOL CORPORATION	0	0	+0	0	0
18	AFLAC INCORPORATED	6	3	−3	3	0
ALL	**Aggregate (10 companies)**	**56**	**64**	**+8**	**10**	1

Consistency check (Net == N→Y − Y→N): **PASS** (+8 == 18 − 10).

A3 (batch 1287) vs A2 (batch 1286)

Company ID	Company	Yes (A2 1286)	Yes (A3 1287)	Net	Y→N	N→Y	T F
9	CSX Corporation	2	5	+3	0	3	3
10	MAGNA INTERNATIONAL INC.	12	12	+0	1	1	2
11	THE SOUTHERN COMPANY	5	3	−2	2	0	2
12	S&P GLOBAL INC.	14	13	−1	4	3	7
13	MARTIN MARIETTA MATERIALS, INC.	0	0	+0	0	0	0
14	DARDEN RESTAURANTS,	0	5	+5	0	5	5

	INC.						
15	BCE INC.	14	16	+2	1	3	4
16	STRYKER CORPORATION	8	7	−1	1	0	1
17	AMPHENOL CORPORATION	0	0	+0	0	0	0
18	AFLAC INCORPORATED	3	3	+0	0	0	0
ALL	**Aggregate (10 companies)**	**58**	**64**	**+6**	**9**	**15**	**2**

Consistency check (Net == N→Y − Y→N): **PASS** (+6 == 15 − 9).

Reading (A3 is shadow-mode — these flips are run variance, not A3 behaviour). A3 deployed with `ELIGIBILITY_ENFORCE` **unset** in prod, so the composition logic attaches an eligibility decision to each `MeasureResult` but mutates **no verdict**; by construction A3 cannot have produced any of the 28/24 flips above — they are the same run-to-run model/corpus variance seen between every pair of runs (cf. A1-vs-base 21 flips, A2-vs-base 20 flips). Direction confirms **no collapse or over-rejection**: the net is a Yes **increase** (+8 / +6), not a loss; the three structurally all-No cos (Martin Marietta, Darden, Amphenol) show zero regression; **Darden 0→5** is a gain, not a collapse; and the single loss — **Aflac −3 vs baseline** — is not new to A3: Aflac dropped 6→3 under A2 as well (see A2 table above) and is **stable A2→A3 (3→3, zero flips)**, i.e. a reproducible corpus artefact independent of this change. The STEP-5 stop-rule (halt on measure-level collapse / over-rejection / regression) is therefore **not triggered**; no second batch was run.

Correctness is established by the unit suite, not the live run. Because the live batch is shadow-mode, the EP01–EP06 guarantees (rejected third-party stays rejected through every path; a subsidiary can't support a parent-wide claim unless scope permits; a current-year question doesn't inherit a superseded report; issuer-filed vs analyst-commentary is distinguished; cache reuse can't carry stale eligibility; incomplete provenance → repair/review, never authority assumption) are proven by the 20 passing tests in `server/lib/eligibility-decision-a3.test.ts`, reproducible with `DATABASE_URL=postgres://stub:stub@127.0.0.1:1/stub npx tsx server/lib/eligibility-decision-a3.test.ts`.

> **GATE3 caveat (no held-out human labels).** There is no adjudicated human-labelled set for this roster, so A3 acceptance rests on (a) the EP01–EP06 unit tests encoding the composition guarantees and (b) demonstrably safe, baseline-stable live behaviour in shadow mode — **not** on measured agreement with human ground truth. Enabling enforcement (`ELIGIBILITY_ENFORCE=on`) should be gated on such a held-out check.

> **Carried-forward constraint (recovery-boundary guard + small-PDF runtime gap).** This open constraint belongs to **A4/WP05** (subprocess PDF extraction — EX04/EX06). A3 touches none of the extraction path (`pipeline.ts` / `processor.ts` are untouched; A3 adds a standalone module, one optional

> `MeasureResult` field, and an env-gated shadow block in `analyzer.ts`), so it does **not intersect** A3 and is **carried forward to A4** unchanged.

Step-1 trace findings (A3 composition gap, confirmed before building)

The plan flagged wrong-entity reinsertion as a *carried-forward* concern to re-verify. Trace confirmed the app already has the individual controls — `classifyProvenance` (`server/lib/provenance.ts`) labels issuer-self / issuer-filed / third-party sources, and `issuer-resolver.ts` derives entity aliases — but there was **no single immutable decision composed from them and carried through scoring**: provenance was classified ad hoc at gate time and not re-pinned per target, so a reused passage or a similarly-named / subsidiary / superseded source had no end-to-end object asserting eligibility. A3 fills exactly that composition gap (compose once → freeze → recompute per target → bounded repair that can't override exclusion), which is why it is a *composition* guardrail over existing controls rather than a replacement for them.

Baseline run (WP01 reference) — ESTABLISHED

- **Batch id:** 1284
- **Deployed commit (all 4 services):** 328d025 (= measurement-foundations code 2b91d86 + docs-only commits; carries stage_profile_events / paid_call_events / run_manifests telemetry sinks)
- **Roster:** frozen 9–18 (CSX, Magna, Southern Co, S&P Global, Martin Marietta, Darden, BCE, Stryker, Amphenol, Aflac)
- **Workspace 3, framework 18 (36 measures), provenance list_id=7**
- **Enqueue:** production-canonical POST /api/admin/resume-analysis (createBatchRun→createAnalysisJobs→addBatchJobs), 10/10 enqueued
- **Pre-run state:** CLEAN SLATE — documents + measure_scores purged, status reset to idle for all 10 (prior state snapshotted to /home/ubuntu/baseline_prerun_snapshot.json: 180 measure_scores, 309 docs). Clean-slate chosen so every company runs full discovery→fetch→extraction→retrieval→scoring (uniform/reproducible starting state + complete stage telemetry + maximal regression coverage); harness-only reset, pipeline/scoring code byte-identical to production.
- **Sole queue consumer:** prod worker (3 replicas); dev worker off shared "analysis" queue (sleep infinity, numReplicas 1).
- **Enqueued at:** 2026-10-15 (resume-analysis HTTP 200)
- **Pass bar (from batch-1275):** 10/10 first-attempt completions, each ≤35 min from claim; any retry voids pass; no accuracy/completeness regression.
- **Status:** ✅ COMPLETE / CAPTURED — 2026-10-15T13:29:51Z.

Baseline captured results (batch 1284) — this is the WP01 reference for all comparisons

- **Completion:** 10/10 completed, 0 failed, **all first-attempt (attempts=1)**. Wall-clock **46m16s** (8-concurrency, 1 prod worker × 3 replicas).
- **Per-company claim→complete:** min 12m29s (Aflac), max 26m42s (Martin Marietta). **NONE exceeded 35 min.** ✅ PASS BAR MET.
  - CSX 23m15s · Magna 20m18s · Southern 20m28s · S&P 26m17s · Martin Marietta 26m42s · Darden 24m43s · BCE 24m05s · Stryker 19m33s · Amphenol 26m16s · Aflac 12m29s.
- **Scores (deterministic reference — keep byte-stable for equivalence WPs):**

id	company	score	met/36	ms
9	CSX	8	3	36
10	Magna	28	10	36
11	Southern Co	6	2	36
12	S&P Global	39	14	36
13	Martin Marietta	0	0	36
14	Darden	0	0	36
15	BCE	39	14	36
16	Stryker	19	7	36
17	Amphenol	0	0	36
18	Aflac	17	6	36

  Aggregate: sum_score=156, sum_met=56, all companies ms=36 (full coverage). Verdict distribution over 360 measure_scores: **No=304, Yes=56**.
- **Cost / telemetry:** total paid LLM cost **$3.9476** (deepseek 1825 calls $3.9464 + openrouter 1 escalation $0.0012); 13,704,064 tokens; 1826 LLM calls, 61 browser (evomi_proxy), 2161 search (serper). 104 stage_profile_events. run_manifest run_id=batch-1284, mode=fresh, hash=fe36c21278c271c6.
- **Interpretation:** three companies scored 0/0 (Martin Marietta, Darden, Amphenol) — all completed with ms=36 (every measure adjudicated), i.e. legitimate “all-No” results, not failures. This all-No cohort is exactly where WP02/A1 research-state accounting (distinguish “looked & found nothing” from “never got doc”) will add signal without changing verdicts.

WP04 / A3 — GATE3 label set (full supporting-quote-set eligibility) — ESTABLISHED

What changed. Commit `9e94507` (“feat(WP04/A3): full supporting-quote-set eligibility basis + quote-correction-primary enforcement”) is on `origin/main` and auto-deployed to all four Railway services (shadow: `ELIGIBILITY_ENFORCE` unset). A3’s eligibility basis now equals the **verdict’s basis** — it evaluates the **entire supporting-quote set** of each positive cell (new pure/total `composeCellEligibility` in `server/lib/eligibility-decision.ts`), not just the single cited quote.

Remedy order is **quote-correction primary, verdict-change last resort**: if any supporting quote is eligible, promote it (`A3-CORRECT`, verdict unchanged); only if **all** quotes are hard-excluded does the verdict downgrade to No/0 (`A3-ENFORCE DOWNGRADE`).

GATE3 forecast — method (replaces the "no held-out labels" caveat with a measured forecast). Rather than a human-labelled set (none exists for this roster), the label set replays the exact enforcement composer over **real production evidence**: batch **1287** snapshot (`analysis_results` id 803, roster 9–18, policy v2). Persisted snapshots drop quote `sourceUrl`, so URLs were **recovered** by matching each quote's `{text, source}` back to the batch corpus (`batch_corpus→documents/document_content`) via NFKC-normalised anchor-window containment. **Recovery rate 99.34% (150/151 supporting quotes; 1 unresolved)** — high enough that the offline replay is rigorous, so **no paid batch was spent** (the fail-silent "URL-drop ⇒ REVIEW" artifact that made the first offline attempt unusable is fully eliminated: **0 cells excluded as artifacts**).

Label buckets (64 positive cells, all currently Yes):

Bucket	Cells	Share	Meaning
STAY	56	87.5%	cited quote already eligible — no change
QUOTE_CORRECT	5	7.8%	eligible supporting quote promoted; **verdict unchanged**
VERDICT_CHANGE	3	4.7%	all quotes hard-excluded → downgrade (last resort)
REVIEW (genuine)	0	0.0%	none eligible, some unknown-provenance
excluded (URL-drop artifact)	0	n/a	—

Directional verdict flips implied by enforcement (per standing reporting constraint):

Direction	Cells	Of which genuine	Of which provenance-attribution artifact
Yes→No	**3**	**0**	**3**
No→Yes	**0**	0	0
Total cells changing verdict	**3**	**0**	**3**

(No→Yes is structurally 0: A3 only acts on positive cells.) Per-company detail and the per-quote exclusion-kind table are in `/home/ubuntu/gate3_labels.md` / `.json`; all 3 Yes→No flips are **S&P Global (co 12)**.

User expectation — CONFIRMED. The governing expectation was that making A3 evaluate the full quote set *"should almost always result in the quote being changed … rather than the verdict being changed."* Among the 8 actionable cells, **quote-correction (5) dominates verdict-change (3)**; VERDICT_CHANGE is 4.7% of classified cells (rare). The data confirms the expectation.

Root-cause of the 3 VERDICT_CHANGE cells — provenance-attribution gaps, not genuine ineligibility. Every excluded quote in the 3 flips is **S&P Global's own disclosure** (2025/2026 Proxy Statements) whose recovered URL the classifier cannot map to the issuer: unmapped **Q4 Inc IR-distribution CDN** tenants (`s29.q4cdn.com`, kind `ir_cdn`) and **search-engine redirect masks** (`www.google.com/goto`, kind `redirect_mask`). These are issuer-self sources mis-scored as third-party — i.e. a classifier provenance gap, **not** evidence the verdict should flip. **Genuine verdict-change after removing the artifacts = 0.**

Enforcement enablement — GATED (two prerequisites, decision required)

Enabling `ELIGIBILITY_ENFORCE=on` **as-is** would apply 3 **false** Yes→No downgrades to S&P Global. That contradicts the data (0 genuine ineligibility) and the project goal ("legitimate results per measure, not fail-silent downgrades"). Enforcement is therefore gated on:

1. **Provenance-gap closure (recommended first; code, not env).** Map the issuer's Q4/IR-CDN tenants into `related_domains` and resolve `www.google.com/goto` redirect masks to their target before provenance classification, so these 3 issuer disclosures score as issuer-self. After this, the forecast becomes **STAY 56 / QUOTE_CORRECT 5 / VERDICT_CHANGE 0** and enforcement produces **zero** verdict flips on this roster — pure quote-correction, exactly the intended behaviour. *(Distinct scope from the approved A3 work; awaiting go-ahead.)*
2. **Production write mechanism.** Flipping `ELIGIBILITY_ENFORCE=on` in prod needs Railway env-var write access (current CLI token here is unauthenticated), **or** approval to change the in-code default to enforce-on (ships via `git push` to main, which is how `9e94507` already deployed; kill switch `ELIGIBILITY_ENFORCE=off` + `ELIGIBILITY_POLICY=v1` remain). Code deploys work from here; **env-var writes do not.**

Carried forward unchanged: WP05/A4 (behavioral recovery-boundary guard + small-PDF runtime gap) remains OUTSTANDING and does not intersect A3.

WP04 / A3 — Framework-level provenance fix SHIPPED + enforcement ENABLED + validated — ESTABLISHED (2026-10-16)

Decision taken. The provenance-gap closure of the section above was executed, and the in-code enforcement default was flipped to **enforce-ON** (both prerequisites resolved). Crucially, the fix was made **framework-level, with zero per-company / per-topic data** — the earlier proposed 1-row `related_domains` tenant declaration for S&P Global (id 12) was **REJECTED by the user as company-specific** and was **never written** (production DB guard also blocked it). It is fully superseded.

Commits on `origin/main` (auto-deploy to all 4 Railway services; dev==prod): `58236a2` (GATE3 label-set docs) → `e06a78f` (enforcement default-ON + redirect-mask unwrap + IR-platform recognition) → **`15bbcc0`** ("feat(provenance+A3): framework-level full-name phrase identity + IR-tenant propagation through eligibility composer") — the superseding framework fix.

What `15bbcc0` ships (all name-derived, company-/topic-agnostic, NO data rows): 1. **Full-name phrase identity** (`server/lib/provenance.ts` `fullNamePhraseMatch`) — an issuer's complete legal name minus generic corporate-form suffixes (Inc/Corp/PLC/Co) is matched as a contiguous phrase against a document's title/URL-path/content, so an issuer with no distinctive single token and no ticker in the URL (e.g. S&P Global, ticker SPGI, on `s29.q4cdn.com/690959130/.../S-P-Global-Inc-2026-Proxy-Statement.pdf`) self-confirms its own IR document. Guarded ≥2 tokens & ≥6 joined chars so short names (AT&T, Visa) do not over-fire. 2. **IR-tenant propagation through the A3 eligibility composer** (`composeCellEligibility`, two-pass, order-independent) — the existing R5c `knownIrTenants` cache (write-through whenever ANY source self-confirms a Q4/IR-CDN tenant) is now threaded through the composer and shared across all of a company's sources (one cache per company in `analyzer.ts` AND in the GATE3 builder, so offline == live). A confirmed tenant then covers sibling documents on the same tenant that carry no self-identifying name. Cross-company reuse guard (`companyMatchesBinding`) still blocks a different issuer borrowing another's bound tenant. 3. **Enforcement default-ON** (`server/lib/analyzer.ts`) — unset/empty `ELIGIBILITY_ENFORCE` now enforces; kill switches intact: `ELIGIBILITY_ENFORCE=off` restores shadow mode, `ELIGIBILITY_POLICY=v1` restores legacy passthrough.

Tests: `eligibility-decision-a3.test.ts` (34) + `provenance.test.ts` (49) pass, incl. 6 new R5c cases (order-independent sibling propagation; cross-company reuse over-fire guard), plus kill-switch and default-ON cases.

Validation — deterministic isolation over intact snapshot 1287 (free, zero-confound)

The fix lives **purely in post-scoring eligibility composition** — it changes neither retrieval nor LLM scoring. The defensible validation is therefore a **deterministic replay of the fixed composer over a fixed snapshot**, which isolates the fix's directional flips with **zero LLM-noise confound**. Snapshot **1287** (analysis_results id 803) is the frozen roster (FW18, companies 9–18, full run) with a **byte-intact corpus** (457/457 docs join, 59.19M chars recovered at 99.34% URL-recovery).

Bucket (64 positive cells, snapshot	Pre-framework-fix	Post-fix (`15bbcc0`,

1287)		enforce-ON)
STAY	56	**63**
QUOTE_CORRECT	5	**1**
VERDICT_CHANGE	3	**0**
REVIEW (genuine)	0	**0**

Directional verdict flips from enabling enforcement (standing reporting constraint): - **Yes→No = 0** (pre-fix would have been **3**, all S&P provenance-attribution artifacts, **0 genuine**) - **No→Yes = 0** (A3 acts only on positive cells) - **Total cells changing verdict = 0**

Only S&P Global (id 12) changed between pre-fix and post-fix; the other 9 companies are **byte-identical — zero regressions**. The one QUOTE_CORRECT (Magna id 10) is the intended remedy (promote an eligible quote, verdict unchanged). Evidence: `deliverables/release-evidence/provenance-gap-closure/AFTER_framework-phrase-and-propagation.{json,md}`.

Why NOT a fresh paid batch — the LLM noise floor swamps the signal

Two fresh runs of the **identical frozen roster with no code change** — baseline **1284** (snapshot 800) vs **1287** (snapshot 803), both run 2026-10-15 — differ by **28 cell flips**:

	Yes (1284)	Yes (1287)	Net	Y→N	N→Y	Total Flips
Aggregate (10 cos)	56	64	+8	**10**	**18**	**28**

This 28-flip run-to-run variance is **pure LLM nondeterminism** (no code differs between the two runs). It is ~10× the fix's isolated effect (0–3 cells), so a fresh paid batch would make the fix's directional flips **unattributable** — the noise would mask the signal. This is why the deterministic composer replay over a fixed snapshot is strictly more defensible for an eligibility-layer change, consistent with the isolation-reliability discipline.

Why baseline 1284 itself could not host the deterministic replay

Baseline **1284**'s underlying corpus **documents have been pruned** from the DB: its `batch_corpus` id-rows survive (333 across 10 companies) but the `batch_corpus → documents → document_content` join returns **0/333** (vs 457/457 intact for 1287). Free URL recovery over 1284 is therefore impossible (measured 0/140 quotes resolved). Snapshot 1287 — the **same frozen roster** — is byte-intact and is the correct validation snapshot. The raw-verdict snapshots for both batches survive (used in the noise-floor table above); only the corpus text for 1284 is gone.

Status: A3 COMPLETE. Framework-level provenance fix shipped (`15bbcc0`), enforcement enabled (default-ON, kill switches intact), validated at VERDICT_CHANGE=0 / Y→N=0 / N→Y=0 with zero

regressions. **Next: WP05/A4** (behavioral recovery-boundary guard + small-PDF subprocess-extraction runtime gap, EX04/EX06).

A4 / WP05 — ESTABLISHED (2026-10-16): extraction / table / OCR / translation fidelity

Scope. Framework-level only — ZERO per-company/per-topic logic or data patches. All change confined to the extraction layer in `server/lib/processor.ts` plus a new test file `server/lib/extraction-fidelity.test.ts`. No schema change, no env-var change, no scoring/analyzer touched. Verdict-invariant by construction.

Files changed - `server/lib/processor.ts` — extraction-engine layer reworked (details below). - `server/lib/extraction-fidelity.test.ts` — new, EX01–EX07 + boilerplate-safety (10 tests). - `docs/implementation-run-ledger.md` — this section + A4 row.

Small-PDF runtime gap — ROOT CAUSE and framework fix. Empirically confirmed: `pdf-parse@1.1.4` bundles an old pdfjs whose xref reader REJECTS minimal hand-built xref tables ("bad XRef entry") — even after the worker's retry-once. A 767-byte fixture fails pdf-parse yet `pdftotext` recovers its full text. The earlier REJECTED approach was a size-special-case bound. The shipped fix is NOT size-based: a pure, exported outcome-classifier `classifyExtractionCompleteness({textLen,minChars,timedOut,failed,pagesAvailable?,pagesProcessed?})` keyed on the *outcome* (parser rejected structure vs. document genuinely empty vs. timed-out partial), not on byte size. "Parser rejected" → `unknown` (retry/review, never a false `empty`); clean run with no text → `empty`; timed-out with partial text → `partial`, never promoted to `complete`. The orchestrator `extractPdfWithFidelity` chains pdf-parse → pdftotext → adaptive OCR and feeds the classifier, so a small PDF rejected by pdf-parse is recovered by pdftotext and labelled `complete`.

Recovery-boundary guard — preserved by non-interference. The behavioural guard lives in `pipeline.ts` (`runCompanyPdfRecovery`, `remainingFetchBudgetMs`, `RECOVERY_WINDOW_FLOOR_MS`) and is proven by `company-pdf-recovery-boundary.test.ts` (B1-1..B1-7). `pipeline.ts` was NOT touched. Confirmed green: **B1-1..B1-7 all pass** post-change.

Subprocess timeout fidelity (EX06). Previously `extractViaPdftotext` silently returned partial stdout on timeout (accepted as success). Now `extractViaPdftotextDetailed` returns `{text, timedOut:true, failed:true}` on timeout; the classifier never promotes timed-out partial stdout to `complete`. `runProcessCollect` still captures partial stdout and flags `timedOut` (verified) — the fix is in how the outcome is *classified*, not in discarding the bytes.

Adaptive OCR (EX02, Lane-B speed gated on Lane-A fidelity). `extractViaOcrDetailed` is per-page: tries the `pdftotext -f p -l p` text layer first and skips OCR for any page already carrying ≥20 chars (text pages preserved verbatim), else rasterises just that page (`pdftoppm`) and OCRs it (`tesseract`). No 8-page cap — all pages processed, budget-bounded; flags `partial` when budget/scope cuts it short (`pagesProcessed < pagesAvailable`). Strictly ⊇ old content.

Table / translation / boilerplate fidelity (f). Asserted against existing framework helpers (not rewritten): `normaliseTableHorizonMarkers` annotates bullet-only horizon rows with

their column relationship; `translation.ts` keeps the aligned derivative (`looksNonEnglish` detects foreign text; translation never replaces the original source carrying numbers/negation/scope); `stripRepeatedStructuralBlocks` dedupes repeated headers but NEVER strips a repeated line containing a quantified figure (policy/decisive note preserved).

Test results (EX01–EX07 + safety) — all green, 10/10 | Test | Result | Proves | |——|———|———-| | EX01 | ok | decisive evidence on a later page survives (no truncation) | | EX02 | ok | mixed doc: text page verbatim AND scanned page recovered by adaptive OCR | | EX03 | ok | table units/headers preserved; horizon row→column relationship annotated | | EX04 (small) | ok | pathological pdf-parse-rejected xref recovered via pdftotext, labelled `complete` (no false empty) | | EX04 (large) | ok | benign 40-page PDF fully extracted, first AND last page markers present (no silent truncation) | | EX05 | ok | translation is an aligned derivative; detector flags foreign text, original source preserved | | EX06 (capture) | ok | `runProcessCollect` captures partial stdout on timeout and flags `timedOut` | | EX06 (rule) | ok | completeness rule NEVER promotes timed-out partial stdout to `complete` | | EX07 | ok | downstream passage recall = 10/10 = 100% across ordinary/multipage/table/OCR/non-Latin | | A4 safety | ok | boilerplate dedupe removes repeated headers but KEEPS a repeated policy line with a figure |

Regression suites — 12/12 green: `company-pdf-recovery-boundary` (B1-1..B1-7, guard preserved), `pdf-parse-subprocess-equivalence` (B6 byte-identical + B3/B2 slot handling), `stalled-extraction-boundary`, `table-horizon-normaliser`.

Typecheck: `tsc --noEmit` — 0 new diagnostics in `processor.ts` / `translation.ts` / `extraction-fidelity.test.ts` (repo has unrelated pre-existing errors in other files).

Cell-flip impact: n/a — no live batch run. The change is verdict-invariant by construction: `extractTextFromPdf` is a thin wrapper returning `ExtractionFidelity.text` byte-for-byte to all 11 existing callers; the fidelity record is additive metadata consumed by nothing in the scoring path. **Total cells changing Y→N = 0, N→Y = 0** (no scoring surface touched).

Boundary noted (not a gap): the `ExtractionFidelity` record (completeness/pages/engine/language/ truncation) is produced and available but not yet threaded into `storage.recordFetchSuccess` / `extraction_state` — intentionally out of scope for this surgical, verdict-invariant change. The `extraction_state` column already supports `partial`; wiring the completeness label through is a follow-up and would remain verdict-invariant.

A5 / WP06 — ESTABLISHED (2026-10-16): retrieval & evidence-pack recall

Scope. Framework-level only — ZERO per-company/per-topic logic or data patches. A valid document can sit in the corpus while its DECISIVE passage never reaches the model prompt; smarter models cannot fix missing input. A5 does NOT add a new retriever — it (a) MEASURES passage-level recall of decisive passages AT THE ACTUAL PROMPT BUDGET, and (b) closes measured misses with a

BOUNDED, append-only evidence-expansion that EXTENDS the existing BM25/deep-read path and COMPOSES with A3 eligibility and A4 fidelity — never by shrinking top-k or pack length.

Files changed - `server/lib/evidence-recall.ts` — new. Pure, DB-free, LLM-free toolkit (oracle + recall + planner). - `server/lib/evidence-recall.test.ts` — new. RT01–RT06 + kill-switch-default + no-op-append (8 tests). - `server/scripts/evidence_recall_benchmark.ts` — new. Free/offline/deterministic recall benchmark. - `server/lib/analyzer.ts` — strictly gated wiring at the pre-scoring evidence seam (details below). - `docs/implementation-run-ledger.md` — this section + A5 row.

Kill-switch and shipped default. `RECALL_EXPANSION_POLICY` (resolved by `resolveRecallExpansionPolicy()`), three values: - **`off` — DEFAULT = current production behaviour.** The analyzer block is skipped ENTIRELY: no extra compute, no extra retrieval, the evidence pack is byte-identical, the verdict surface is unchanged. - `shadow` — compute recall telemetry + the expansion plan but DO NOT modify the pack (verdict-invariant; observability only). - `on` — APPLY the bounded expansion (behaviour-changing; append-only). Any unrecognised value falls back to `off` (fail-safe — a typo can never silently enable expansion). Verdict-changing retrieval is NEVER the default and is NOT set in the prod env. **To enable: `RECALL_EXPANSION_POLICY=on`** (or `=shadow` for telemetry); unsetting it restores current behaviour with no redeploy.

Decisive-passage oracle (framework-level). `computeDecisivePassages({measure,chunks})` marks a passage decisive when it carries ≥ 2 distinct decisive tokens OR a verbatim multi-word `evidenceKeyword` phrase. The decisive signal is drawn ONLY from the measure's own framework-authored fields (`title` + `evidenceKeywords`) — no company names, no topic hardcoding. Ranked (phraseHit, distinctTokens, stable id) and capped (`RECALL_DECISIVE_MAX`, default 12), so it is deterministic and bounded. This single definition is BOTH the recall ground-truth AND the candidate universe for expansion, so "what should have reached the model" and "what expansion may add" can never diverge.

Recall measurement at the actual prompt budget. `measurePackRecall({decisive,packPassageIds})` reports |decisive ∩ pack| / |decisive| by STABLE passage identity (`stablePassageId` = `docUrl#seq#sha1(text)`, mirrors the evidence-pack fingerprint scheme), plus the exact misses in oracle order. `packPassageIds` are the ids of the chunks that ACTUALLY reached the pack shown to the scoring role (membership recovered by verbatim substring against the assembled pack text).

Bounded expansion planner (A3-composing, append-only, deterministic). `planBoundedExpansion(...)` fires only on DEMONSTRATED insufficiency (an eligible decisive passage exists in the corpus but missed the pack). It: - COMPOSES with A3: an injected eligibility predicate filters excluded material BEFORE selection, so expansion can NEVER reintroduce what A3 excluded (RT04). The analyzer injects the real A3 predicate (`composeEligibilityDecision` per candidate; `status==="excluded"` ⇒ ineligible), fail-open on error. - is BOUNDED: ≤ `RECALL_EXPANSION_MAX_PASSAGES` (default 6) and ≤ `RECALL_EXPANSION_MAX_CHARS` (default 12000) per measure — no unlimited retrieval. - NEVER shrinks top-k or pack length (append-only; the original pack is intact, order preserved). - dedups ONLY when a candidate adds no new decisive token AND its source link is

already represented — so every unique decisive fact AND source link stays recoverable; qualifiers/negation in a distinct passage are never dropped (RT06). - is DETERMINISTIC: stable identity + stable ordering, re-runs byte-identical (RT05).

analyzer.ts wiring (strictly gated). At the pre-scoring seam (`finalEvidence = evidenceText`), guarded by `if (recallPolicy !== "off" && settings.useBm25Retrieval && evidenceText)`. When `off` the whole block is skipped (no `buildDeepIndex()`, no oracle, no plan) ⇒ zero added cost and byte-identical packs. When `shadow` it logs recall telemetry only; when `on` it appends `formatExpansionAppend(plan)` (a provenance- headed block) to `finalEvidence`. No existing pack content is removed or reordered.

Deterministic test results (RT01–RT06 + extras) — 8/8 pass | Test | Result | Proves | |——|———–|———–| | RT01 | ok | synonym/sector-language passage is decisive and recalled at the budget | | RT02 | ok | a low-frequency phrase-hit (table cell) is decisive and recoverable by bounded expansion | | RT03 | ok | negative/contradictory evidence is decisive and never suppressed | | RT04 | ok | bounded expansion composes with A3 — excluded material is never reintroduced | | RT05 | ok | stable passage identity and deterministic ordering under input shuffling | | RT06 | ok | dedup drops only pure duplicates; unique facts/links/negation preserved | | kill-switch | ok | default is `off` (current production behaviour, verdict-invariant) | | no-op append | ok | `formatExpansionAppend` returns `""` for a non-firing plan (byte-identical pack) |

Run: `DATABASE_URL=postgres://stub:stub@127.0.0.1:1/stub npx tsx server/lib/evidence-recall.test.ts` (the tests never touch the DB; the stub only satisfies the pool constructor).

GATE2 — free/offline/deterministic recall benchmark. `server/scripts/evidence_recall_benchmark.ts` builds the EXACT pack the scorer sees using the REAL, deterministic `summarizeDocuments` (retrievalV2 BM25 corpus selection — NO LLM) + the REAL `buildEvidencePacksForCategory`, over the frozen fixture corpus (`var_corpus/*`), and reports passage-level recall + shadow recoverability. Read-only w.r.t. the production DB (needs only an empty throwaway `summary_cache`; a cache miss is the normal path). Full 13-company fixture roster, all 29 measures, 0 company errors:

Metric	Value
Companies evaluated	13 (ids 1, 20, 120, 22, 107, 21, 14, 18, 11, 12, 13, 102, 16)
(company × measure) cells	377
Decisive passages (corpus total)	4493
Recalled into the budgeted pack	3334
Micro-averaged recall	**74.2%**
Macro-averaged recall	74.3%
Decisive misses	1159
Recoverable by A5 bounded expansion (shadow)	537 (46.3% of misses)
Budget	corpusMaxChars 560000, top-k 45, evidenceMaxChars 45000
Resolved policy during run	`off` (expansion run in SHADOW for measurement; pack never

modified)

Outputs: `var_out/evidence_recall_benchmark.{json,md}` (per-cell + aggregate). Re-runs are byte-identical.

Typecheck: `tsc --noEmit` — 0 new diagnostics in `evidence-recall.ts` / `evidence-recall.test.ts` / `evidence_recall_benchmark.ts` / `analyzer.ts` (repo has 62 unrelated pre-existing errors in other scripts/files).

Adjacent regression suites — all green: evidence-gate 14/14, evidence-gate-a2 17/17, eligibility-decision-a3 34/34, passage-retrieval 35/35, retrieval-stability 5/5, retrieval-review-queue (green). No regression in the retrieval/evidence-gate/eligibility surfaces A5 sits next to.

Cell-flip impact: n/a — no live batch run, and none is required. The shipped default `RECALL_EXPANSION_POLICY=off` skips the analyzer block entirely, so the evidence pack and verdict surface are byte-identical to current production. **Total cells changing Y→N = 0, N→Y = 0** (no scoring surface touched at the default). Behaviour-changing recall (`on`) is opt-in and NOT enabled in prod.